CM3121 – Advanced Organic Chemistry

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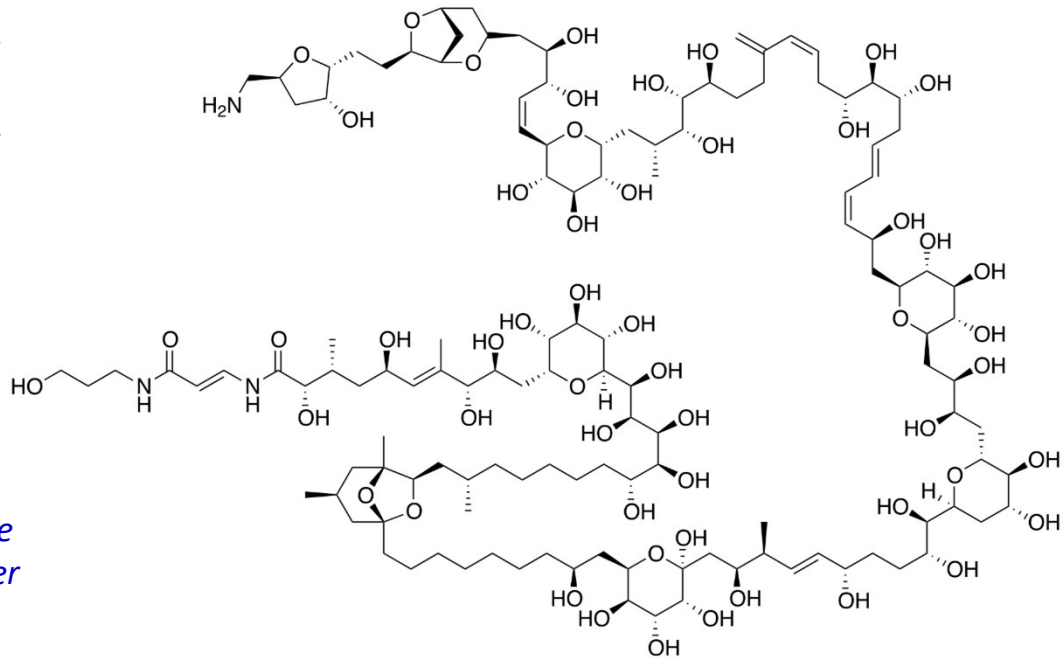
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Protecting Groups

Readings: Clayden 2nd ed Chapter 23 (Protecting groups)

Target molecules

Palytoxin is a polyhydroxylated and partially unsaturated compound (8 double bonds) with a long carbon chain. It has water-soluble and fat-soluble parts, 40 hydroxy groups and 64 chiral centers. Due to chirality and possible double bond cis-trans isomerism, it has over 10^{21} alternative stereoisomers.

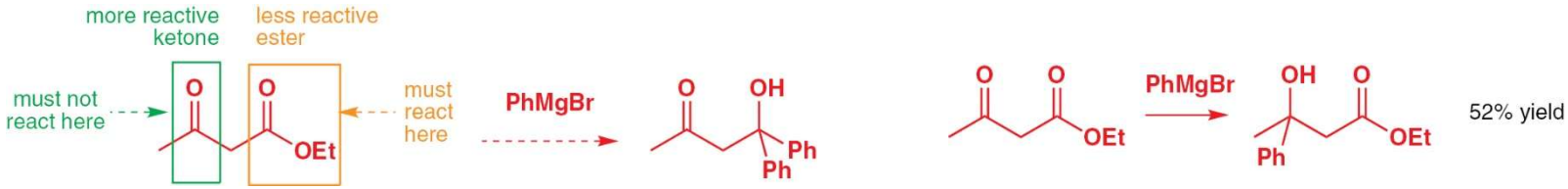


"the Mount Everest of organic synthesis, the largest single molecule that anyone has ever even thought about making"

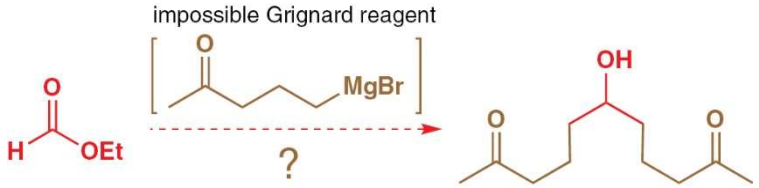
https://en.wikipedia.org/wiki/Palytoxin#References

When to use protecting groups?

1. Two (or more) functional groups and you need to react the less reactive functional group (or one of same functional group). Examples: electrophilicity: ketone >> ester

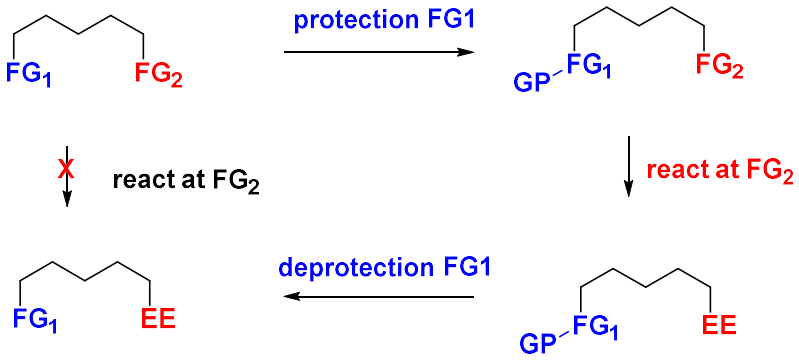


2. Incompatible functional groups.
Examples: - Grignard reagent (nucleophile) vs carbonyl group (electrophile) or
- H^+ sources, such as carboxylic acid (RCOOH) or alcohol (ROH) *and Grignard reagent*



How to use protecting groups?

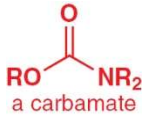
Solution: protecting group



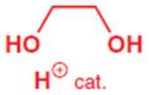
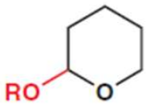
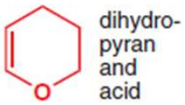
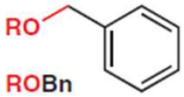
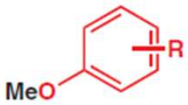
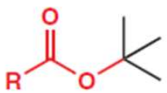
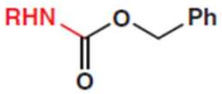
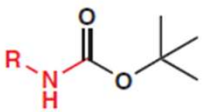
- *Protection and deprotection reactions*
- *Stability of the protecting groups*

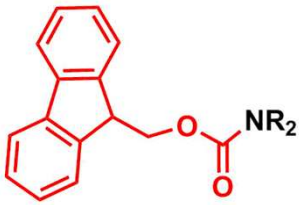
Common protecting group:

- Alcohols: silyl ethers, benzyl ethers
- Amines: carbamates



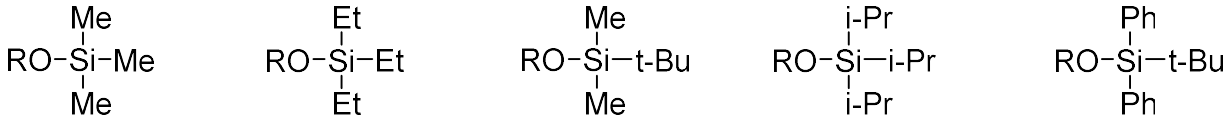
- Aldehydes and ketones: acetals
- Carboxylic acids: esters

Protecting group	Structure	Protects	From	To protect	To deprotect
acetal (dioxolane))		ketones, aldehydes	nucleophiles, bases		H^+, H_2O
trialkylsilyl R_3Si (e.g.TBDMS)	RO–$SiMe_3$ RO–$SiMe_2$*t*-Bu	alcohols (OH in general)	nucleophiles, C or N bases	R_3SiCl, base	H^+, H_2O, or F^-
tetrahydropyranyl (THP)		alcohols (OH in general)	strong bases		H^+, H_2O
benzyl ether (OBn)		alcohols (OH in general)	almost everything	NaH, BnBr	H_2, Pd/C, or HBr
methyl ether (ArOMe)		phenols (ArOH)	bases	NaH, MeI, or $(MeO)_2SO_2$	BBr_3, HBr, HI, Me_3SiI
t-butyl ester (CO_2*t*-Bu)		carboxylic acids	nucleophiles	isobutene, H^+	strong acid
Cbz (Z) (OCOBn)		amines	electrophiles	BnOCOCl, base	HBr, AcOH; or H_2, Pd
t-Boc (OCO*t*-Bu)		amines	electrophiles	$(t\text{-BuOCO})_2O$, base	H^+, H_2O
Fmoc	see text	amines	electrophiles	Fmoc-Cl	base, e.g. amine



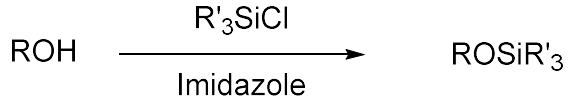
a. Protecting groups for alcohols: silyl ethers

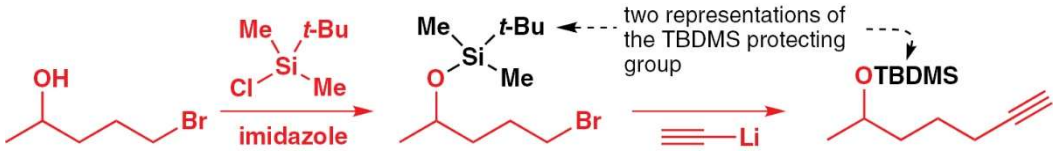
Common silyl ether protecting groups



TMS TES TBS TIPS TBDPS

Protection:

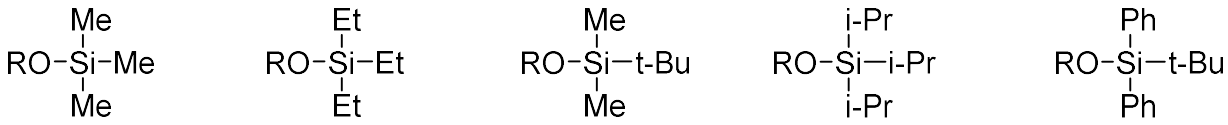




Deprotection:

1. Tetrabutylammonium fluoride (TBAF): nBu$_4$NF, or
2. Acid and water

Silyl ethers



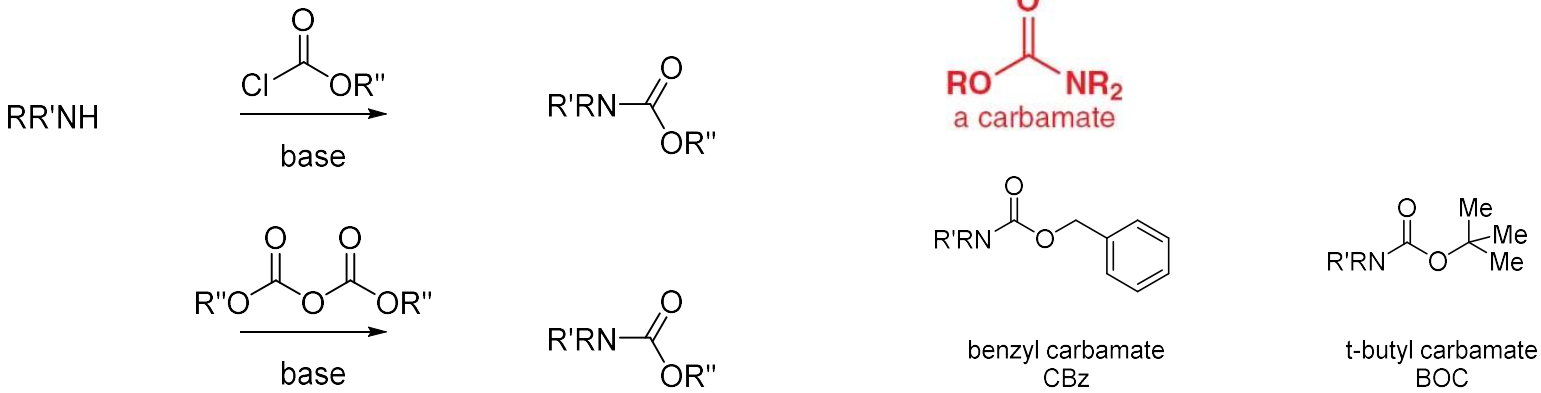
TMS TES TBS TIPS TBDPS

Stability toward acidic media:
TMS (1) < TES (64) < TBS (20,000) < TIPS (700,000) < TBDPS (5,000,000)

Stability towards basic media:
TMS (1) < TES (10-100) < TBS ~ TBDPS (20,000) < TIPS (100,000)

b. Protecting groups for amines

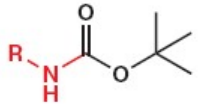
Protection:

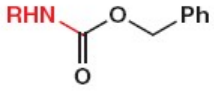


Deprotection:

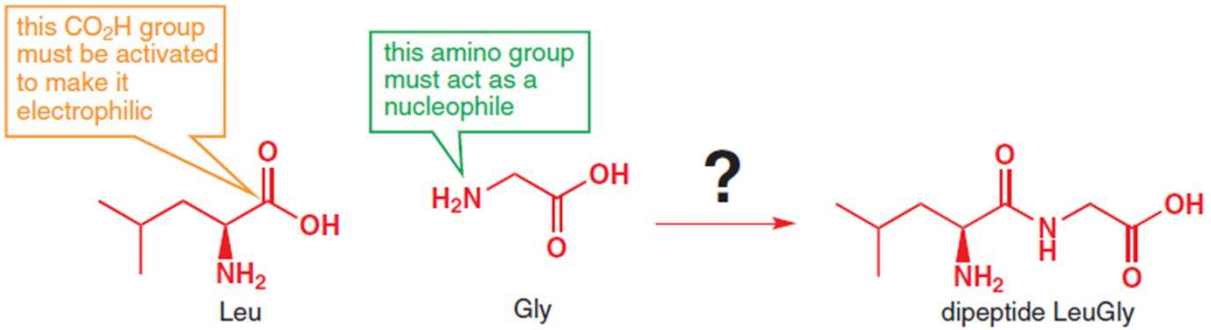
Boc: acid (e.g., TFA)

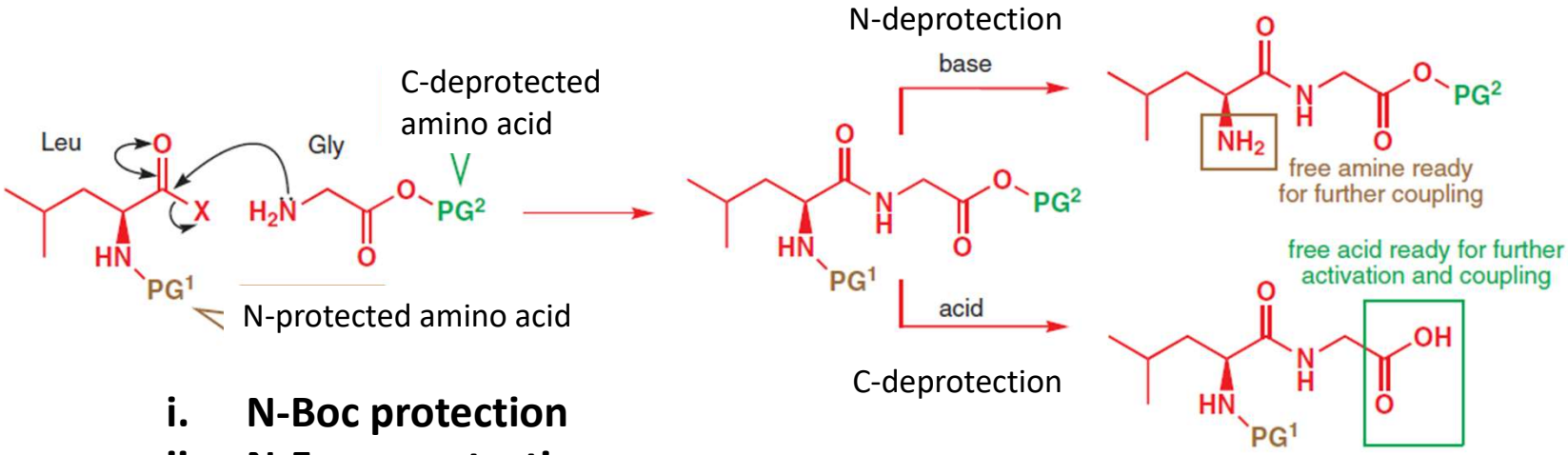
CBz: H_2/Pd (or acid)

Protecting group	Structure	Protects	From	To protect	To deprotect
Boc (*t*-BuOCO)		amines	electrophiles	$(t\text{-BuOCO})_2O$, base	H^+, H_2O

Protecting group	Structure	Protects	From	Protection	Deprotection
Cbz (Z) (OCOBn)		amines	electrophiles	BnOCOCl, base	HBr, AcOH; or H_2, Pd

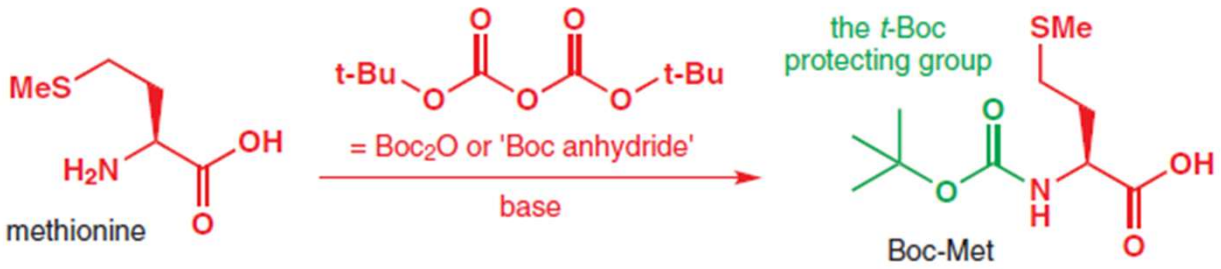
Peptide synthesis

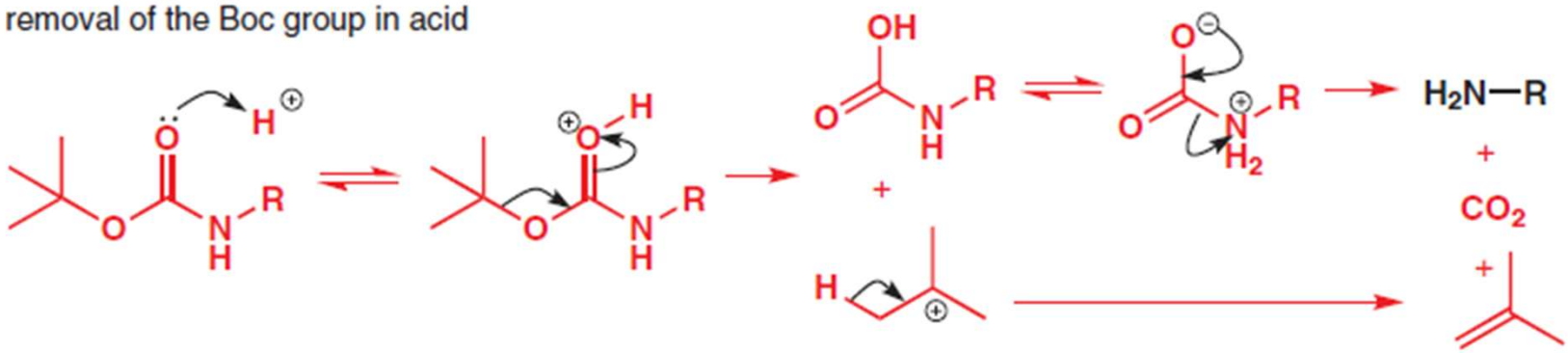




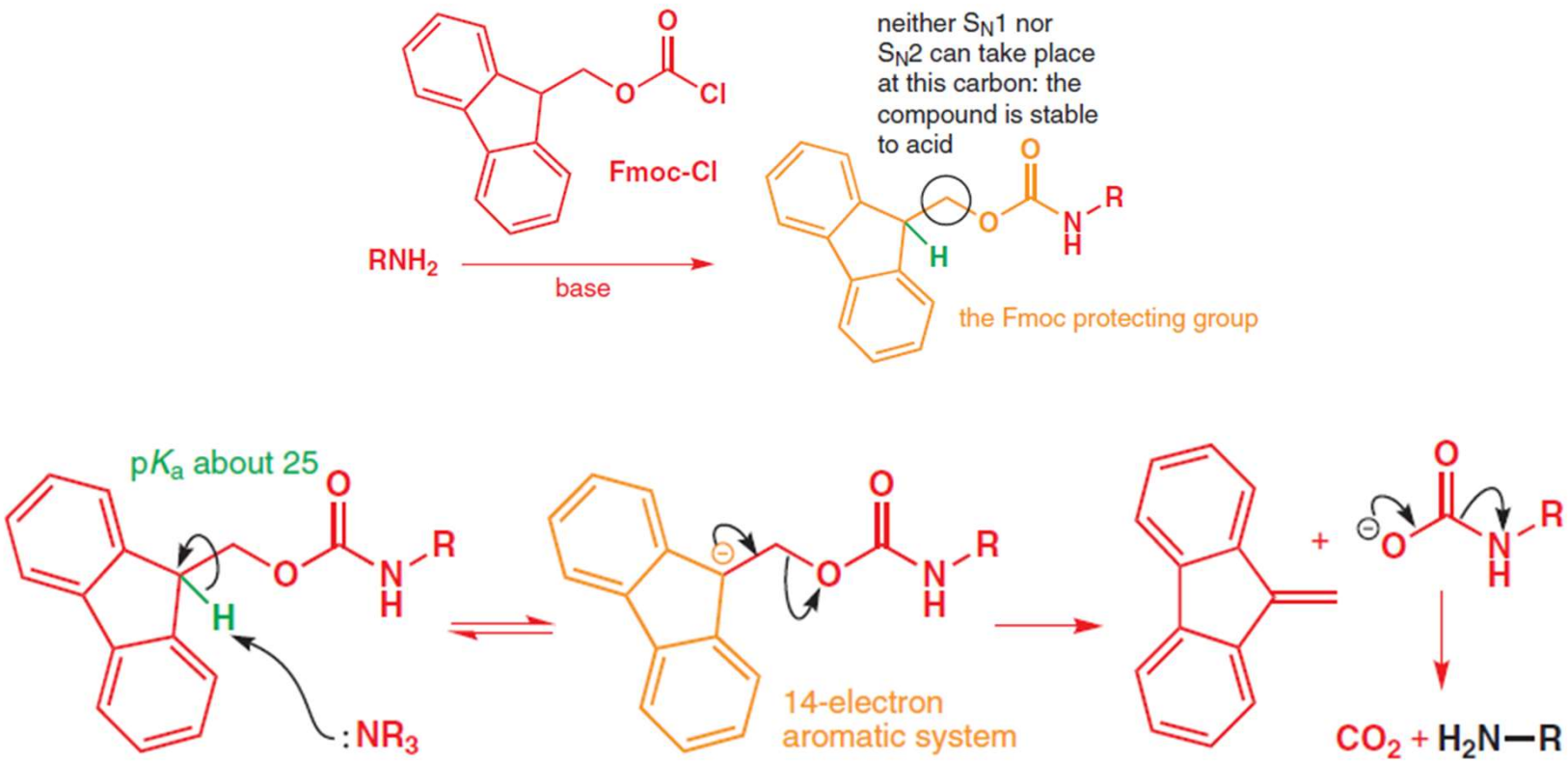
i. **N-Boc protection**
ii. **N-Fmoc protection**

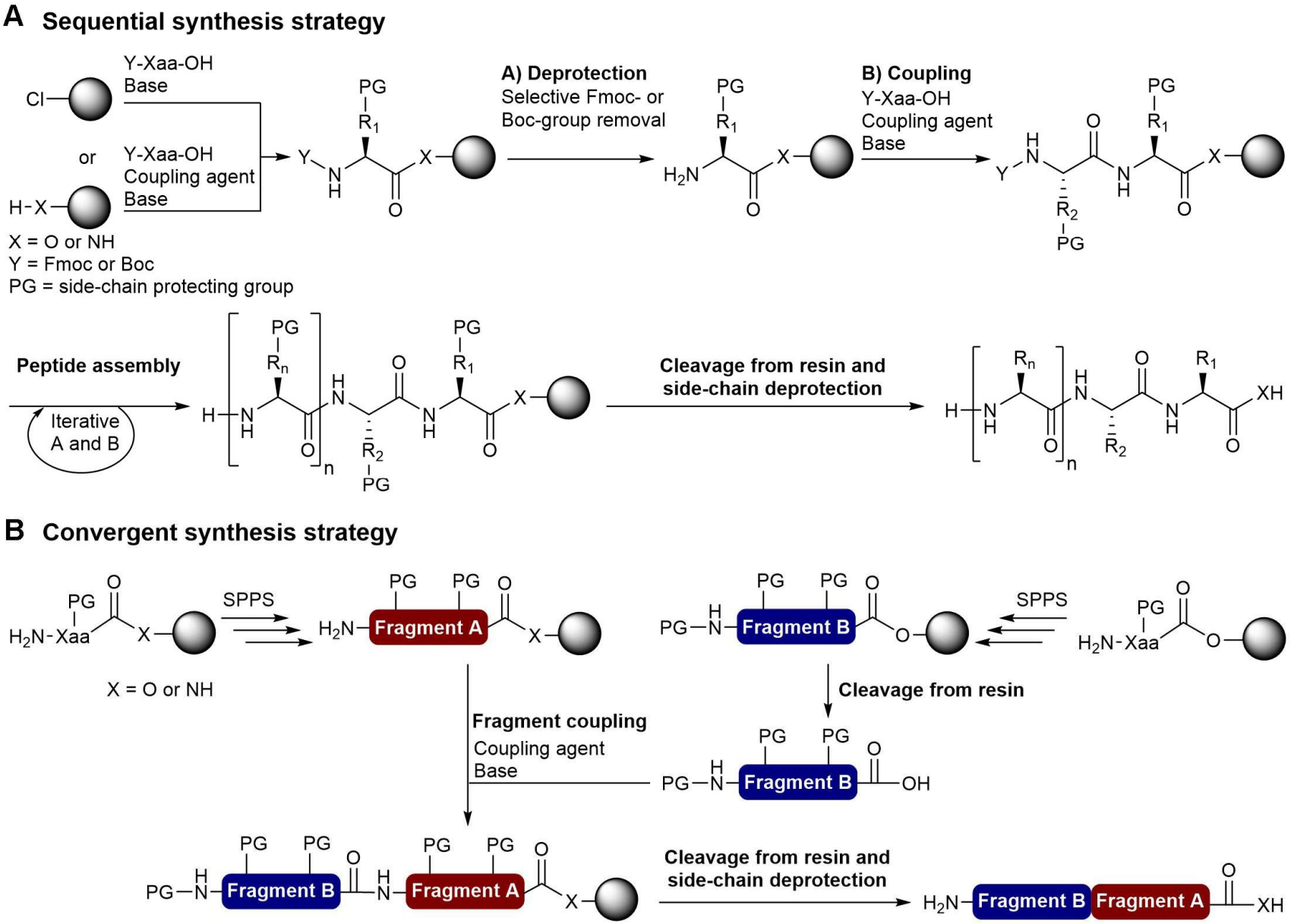
i. N-Boc protection





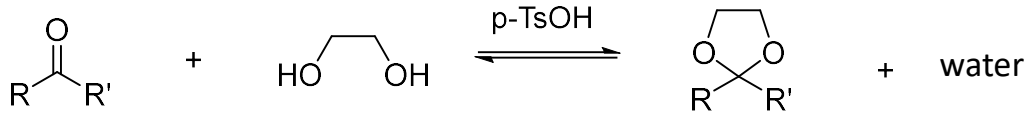
ii. N-Fmoc protection



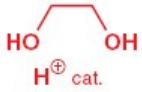

A Sequential synthesis strategy
Y-Xaa-OH
Base
or
Y-Xaa-OH
Coupling agent
Base
X = O or NH
Y = Fmoc or Boc
PG = side-chain protecting group
A) Deprotection
Selective Fmoc- or
Boc-group removal
B) Coupling
Y-Xaa-OH
Coupling agent
Base
Peptide assembly
Iterative
A and B
Cleavage from resin and
side-chain deprotection
B Convergent synthesis strategy
SPPS
Fragment A
Fragment B
X = O or NH
Cleavage from resin
Fragment coupling
Coupling agent
Base
Cleavage from resin and
side-chain deprotection

c. Protecting groups for aldehydes or ketones

Protection: as acetal form (ethylene glycol, p-TsOH)



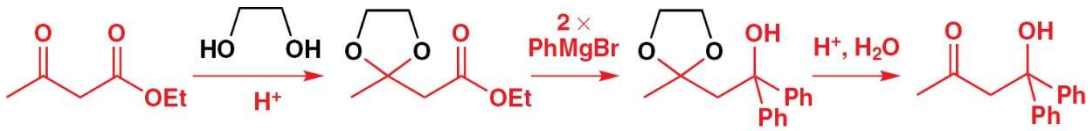
Acetals are stable to bases or nucleophiles but not stable under acidic conditions.
Deprotection: H_2O, p-TsOH

Protecting group	Structure	Protects	From	To protect	To deprotect
acetal (dioxolane)		ketones, aldehydes	nucleophiles, bases		H^+, H_2O

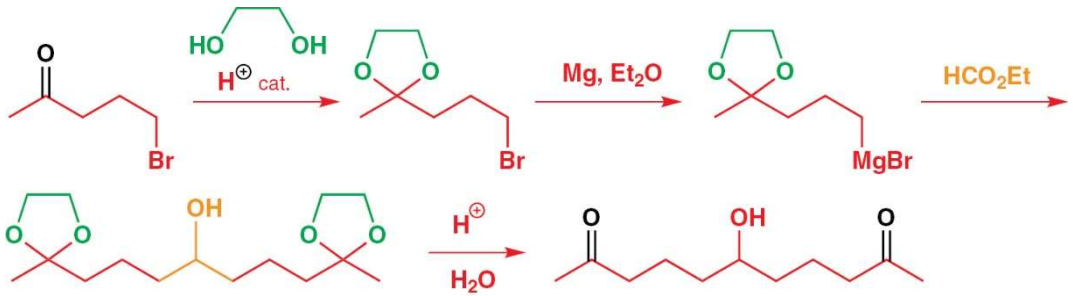
Protecting groups for aldehydes or ketones

Examples

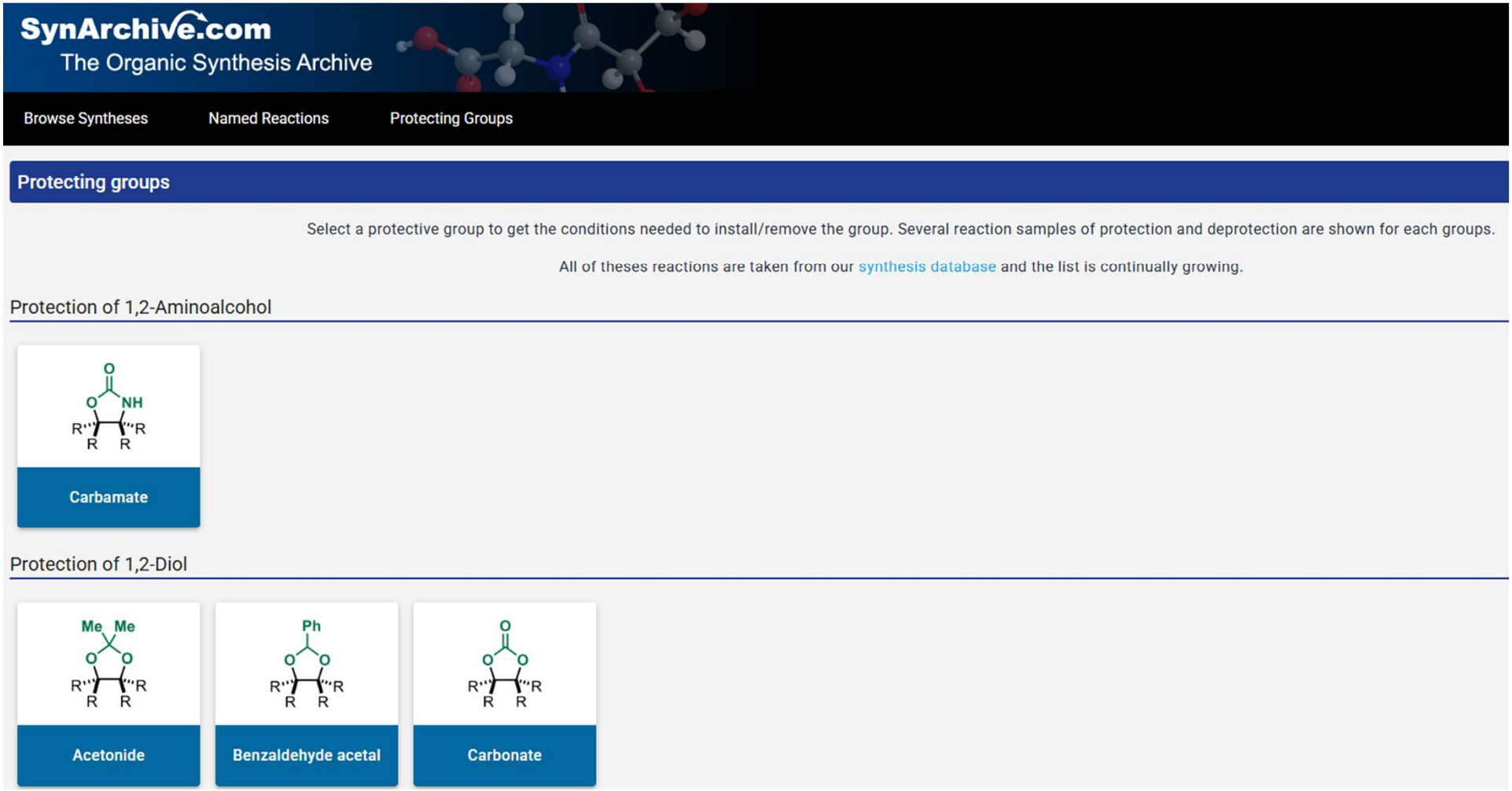
1. Two (or more) functional groups and you need to react the less reactive functional group (or one of same functional group).
   Examples: electrophilicity: ketone >> ester



2. Incompatible functional groups.
   Examples: - Grignard reagent (nucleophile) vs carbonyl group (electrophile) or
   - H^+ sources, such as carboxylic acid (RCOOH) or alcohol (ROH) *and Grignard reagent*



Information on protecting groups



https://synarchive.com/protecting-group

Summary

Protecting groups are temporary “masks” used to block a functional group from reacting while other transformations are performed elsewhere in the molecule.

FG1: Incompatible

Protection of FG1 → Reaction at FG2 → Deprotection of FG1

- Alcohols: silyl ethers
- Amines: carbamates
- Aldehydes and ketones: acetals

